#ManyDaughters - Pre-Analysis Plan

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1 Introduction

This study is part of the #ManyDaughters research initiative, which aims to examine whether having daughters causally influences individuals' political preferences, gender attitudes, and gender-related household behaviors. We use data from the German Socio-Economic Panel (SOEP), a nationally representative longitudinal survey of private households in Germany (Goebel et al., 2019), as supplied by the #ManyDaughters team. This pre-analysis plan evaluates four hypotheses concerning the relationship between child gender and adult attitudes or behavior:

H_1 Having daughters leads to stronger support for gender equality, as measured by attitudinal responses.
Let X denote a continuous measure of support for gender equality. Then:

$$\mathbb{E}[X \mid \text{has daughters}] > \mathbb{E}[X \mid \text{no daughters}]$$

H_2 Having daughters leads to more left-leaning political preferences.
Let Y denote a measure of left-leaning political preferences. Then:

$$\mathbb{E}[Y \mid \text{has daughters}] > \mathbb{E}[Y \mid \text{no daughters}]$$

H_3 Having daughters leads to more gender equality in the household, as measured by behaviors or outcomes.
Let Z denote a measure of gender equality in household behavior. Then:

$$\mathbb{E}[Z \mid \text{has daughters}] > \mathbb{E}[Z \mid \text{no daughters}]$$

H_4 The effect of having daughters on support for gender equality is greater for men than for women.
Let X again denote a measure of support for gender equality. Then:

$$\begin{aligned}&(\mathbb{E}[X \mid \text{male, has daughters}] - \mathbb{E}[X \mid \text{male, no daughters}]) \\ &> (\mathbb{E}[X \mid \text{female, has daughters}] - \mathbb{E}[X \mid \text{female, no daughters}])\end{aligned}$$

2 Sample

We use data from the German Socio-Economic Panel (SOEP), a nationally representative longitudinal household survey of private households in Germany, managed by DIW Berlin. Specifically, we rely on harmonized SOEP-Core data compiled for the #ManyDaughters project, covering the years 2000 to 2020. The dataset includes panel information from all individuals aged 18 and over who completed at least one wave of the individual questionnaire during this period (N $\approx$ 90,000). We construct separate analytical samples for individual- and household-level hypotheses, reflecting differences in outcome definition and required data structures.

Individual-Level Hypotheses (H_1, H_2, H_4). For the hypotheses related to attitudinal and political outcomes, we restrict the sample to:

- Biological mothers and fathers whose first-born biological child has known and binary-coded gender (male or female).
- Individuals with valid data on at least one of the outcome variables and complete data on all covariates required for the respective regression models.

For the sake of simplicity, we exclude individuals who are not biological parents, those with children of unknown or non-binary gender, and those with missing covariate information. Respondents are retained even if they provide outcome data in only one wave, as long as the required covariates are available.

Household-Level Hypothesis (H_3). For the hypothesis which concerns intra-household behavior and outcomes, we focus on a subsample of households with defined parental dyads. Specifically, we restrict the sample to:

- Cohabiting heterosexual couples in which both adults are the *biological parents* of a first-born child of father and/or mother.
- Both parents must be active respondents in at least one wave in which relevant behavioral outcomes are observed.
- The first-born child has known, binary-coded gender.
- Non-missing data on at least one outcome and all control variables.

This restriction enhances comparability by standardizing family structure, biological parentage, and shared household context, thereby reducing confounding from variation in co-residence, step-parenting, or household composition. Households with step-parents, single parents, or same-gender adult dyads are excluded from the main analysis. Similarly, households with incomplete outcome or covariate data, missing child gender information, or non-cohabiting biological parents are also excluded.

To construct matched within-household measures, we link co-residing partners using household identifiers (`hhid`) and person identifiers (`pid`) via the SOEP long

format linking file (`ppathl`) and the household roster (`hgen`). The resulting matched dataset allows us to analyze the actual behavior and outcome by gender, and to assess whether having daughters predicts more equitable sharing.

2.1 Sample Construction

All samples are constructed from harmonized SOEP-Core modules, including `pl`, `hl`, `biol`, `biobirth`, `bioparen`, `pgen`, `hgen`, and `ppathl`. Data construction and linkage are performed in Stata 14 or higher and fully documented for replication. Final sample sizes vary by outcome availability, respondent coverage, and panel structure.

Sample 1. Our primary identification strategy relies on a difference-in-differences framework. We focus on respondents whose first biological child was born during the sampling window (2000-2020) and who are observed at least once both before and after the birth of that child. This subset allows us to exploit within-individual variation in daughter exposure while conditioning on time-invariant unobserved heterogeneity. For these respondents, all observations prior to treatment (pre-birth) are classified as control, and those following treatment onset (post-birth) as treated. Only individuals or households with complete covariate data and at least one valid outcome measurement in both periods are retained.

Sample 2. If these constraints prove too restrictive – for instance, if resulting sample sizes fall below the pre-specified analytic threshold – we expand the sample to include all respondents with known and binary-coded first child gender, irrespective of whether they are observed pre- and post-birth. In this case, we treat child gender as a time-invariant treatment and estimate cross-sectional or pooled models, controlling for observed covariates and fixed effects. This fallback strategy ensures empirical feasibility while maintaining consistency with the overall design logic.

To avoid over-representation of individuals with longer or more frequent panel participation, we retain at most one observation per individual per treatment status. For respondents observed in multiple years under the same treatment condition (e.g., prior to or following the birth of a first-born daughter), we retain the last available observation, conditional on complete data for the outcome and all required covariates. For those who do not experience a status change during the observation window, a single final observation is retained. This sampling rule prioritizes later-period responses, under the assumption that behavioral and attitudinal effects of having a daughter may evolve over time and stabilize with exposure. While this approach may increase sensitivity to attrition-related selection, it enables consistent estimation of long-run effects across individuals with varying panel lengths.

3 Variables

This section describes the construction and coding of the treatment, outcome, and control variables used in the empirical analyses. All variables are constructed from harmonized SOEP-Core datasets provided for the #ManyDaughters project, primarily based on modules `pl`, `hl`, `biol`, `biobirth`, `bioparen`, `pgen`, `hgen`, and `ppathl`.

3.1 Treatment Variable

The main treatment of interest is exposure to biological daughters. To strengthen identification and reduce bias, we restrict all analyses to families where the observed child is the *first-born*. This variable, `first_child_female`, equals 1 if the respondent's first-born biological child is a daughter and 0 if the first-born is a son.[1] The sex of the first child is obtained from the SOEP parent–child linkage files (`biol`, `biobirth`).

We focus on first-born children to mitigate confounding from fertility-related selection mechanisms, including gender-stopping behavior and selective parity progression. This restriction also enhances causal interpretability, as the sex of the first child is plausibly exogenous in the German context during the observation window. Moreover, theoretical and empirical work suggests that first-born children may disproportionately shape parental attitudes and household norms. Alternative exposure measures (e.g., ever having a daughter, proportion of daughters) are not employed in this study to maintain a unified identification framework across hypotheses and analyses.

All analyses consistently use `first_child_female` as the treatment indicator, with respondents whose first-born is male serving as the control group.

3.2 Outcome Variable Description

To test the hypotheses outlined, we construct outcome variables in three domains: support for gender equality (section 3.2.1), political preferences (section 3.2.2), and gender equality within the household (section 3.2.3). Each domain includes two complementary measures.

3.2.1 Support for Gender Equality (H_1, H_4)

We construct two measures of support for gender equality. The first measure is an index based on five attitudinal items (`plj0620` to `plj0625`, excluding `plj0624`), which capture views on gender roles in education, employment, and family life. These items are available in the 2016, 2017, and 2018 waves and are framed such that higher values indicate more egalitarian attitudes. The index is computed as the mean of the non-missing item responses. Respondents with fewer than three valid item responses are coded as missing for the index. Item wording and scale orientation are consistent across waves.

The second measure is the single item `plh0308`, which captures agreement with the statement: "It is best if men and women work the same amount so they can share the responsibility." This item is available in 2012 (4-point scale) and 2018 (7-point scale). To harmonize across years, we apply min–max rescaling to map all responses onto the unit interval $[0, 1]$, preserving within-wave relative positioning

[1] In case of non-identical multi-child births with at least one daughter, the family is counted in the group of daughters.

while accommodating differences in response scale granularity. Higher values indicate stronger agreement with egalitarian sharing norms.

Both measures are analyzed separately and treated as continuous outcomes. In both specifications, the attitudinal measures are interpreted as proxies for support for gender equality in values rather than behaviors.

To comply with the specifications of the project, in the end we only consider one measure for each output file, i.e. only the first if we have enough observations, and only the second if we do not have enough observations for measure one.

3.2.2 Political Preferences (H_2)

We measure political preferences using two indicators. The first is the respondent's left-right self-placement (`plh0004`), recorded on an ordinal scale from 0 (left) to 10 (right), and available in the 2005, 2009, 2014, and 2019 waves. Following standard practice in applied work, this scale is treated as continuous in the main specification. We exclude respondents who do not place themselves on the scale or select "don't know."

The second indicator is party identification (`plh0012_h`), available in all survey years. Responses are recoded onto a standardized left-right dimension using the time-consistent classification provided by Jolly et al. (2022). This mapping assigns numerical ideological positions to all major parties represented in the Bundestag, enabling the construction of a respondent-level left-right score based on party preference. Respondents who indicate no party affiliation or refuse to answer are coded as missing.

Both variables are analyzed separately and treated as continuous outcomes.

To comply with the specifications of the project, in the end we only consider one measure for each output file, i.e. only the first if we have enough observations, and only the second if we do not have enough observations for measure one.

3.2.3 Gender Equality in the Household (H_3)

We construct two measures of male participation in household responsibilities as indicators of gender equality in the household. The first outcome is the male partner's share of total weekday hours spent on housework and childcare, based on responses to `pli0043_h` and `pli0044_h`. Higher values on these measures reflect greater male participation in domestic labor and are interpreted as indicative of more gender-equal intra-household arrangements.

The second is the male partner's share of net monthly household income, calculated from `plc0014_h`. Lower values are interpreted as indicative of more gender-equal intra-household arrangements.

Both outcomes are expressed as proportions ranging from 0 to 1.

To comply with the specifications of the project, in the end we only consider one measure for each output file, i.e. only the first if we have enough observations, and only the second if we do not have enough observations for measure one.

3.3 Control Variables

Ideally, regression specifications without individual fixed effects (specification 2) include a standard set of covariates that account for demographic composition, socioeconomic status, and geographic location. These are motivated by prior research on family dynamics, political preferences, and gender attitudes. Subject to availability, we aim to include:

- **Age**: Computed as interview year minus year of birth (`gebjahr`).
- **Gender**: Binary indicator, 1 = female, 0 = male (`pla0009_h`).
- **Number of Children**: Count of biological children present at the time of interview
- **Children Composition**: Proportion of daughters.
- **Education**: For individual-level hypotheses, highest degree or harmonized years of schooling (e.g., ISCED or `pgbilzeit`). For household-level hypothesis, highest degree or harmonized years of schooling (e.g., ISCED or `pgbilzeit`) of the father as well as difference to mother's education.
- **Household Income**: Monthly net income, equivalized and log-transformed.
- **Urbanicity**: Urban vs. rural residence indicator. (If `regtyp` Spatial category by BBSR is available in the data set)
- **Household Size**: Number of individuals residing in the household.
- **Federal State**: State fixed effects.
- **Survey Wave**: Year fixed effects.

4 Empirical Strategy

Specification 1. To estimate the effect of having daughters on political preferences, gender attitudes, and intra-household behaviors, we adopt a difference-in-differences (DiD) strategy as our primary empirical approach. This design exploits within-individual variation by comparing outcomes before and after the birth of the first child, conditional on the gender of that child. The analysis is restricted to respondents whose first biological child is born within the observation window (2000-2020) and who are observed at least once before and once after the child's birth. This restriction enables credible identification of treatment effects under the assumption that, absent treatment, potential outcomes would have followed parallel trends. That is, we estimate individual fixed effects models of the form:

$$Y_{it} = \beta_1 \mathrm{B}_{it} \times T_i + \beta_2 \mathrm{B}_{it} + \gamma X_{it} + \delta_t + \alpha_i + \epsilon_{it} \tag{1}$$

where Y_{it} denotes the outcome of interest for individual i in year t, B_{it} is a binary variable, marking observations as pre (0) or post birth (1), T_i is a binary indicator equal to one if the respondent has or will have a first-born child who is female, X_{it} is a vector of potentially time-varying covariates, δ_t denotes survey year fixed effects, and α_i captures individual fixed effects. The coefficient β_1 identifies the average within-person effect of having a first-born daughter.

To test H_4, which posits heterogeneous effects by respondent gender, we extend the model with an interaction term:

$$Y_{it} = \beta_1 \mathrm{B}_{it} \times T_i \times \mathrm{m}_i + \beta_2 \mathrm{B}_{it} \times T_i + \beta_3 \mathrm{B}_{it} \times \mathrm{m}_i + \beta_4 \mathrm{B}_{it} + \gamma X_{it} + \delta_t + \alpha_i + \epsilon_{it} \quad (2)$$

where m_i is a binary indicator equal to one for male respondents, and the remaining variables are specified as stated above.

Specification 2. When the DiD design is not feasible, we estimate pooled cross-sectional models including a set of available covariates and include survey year and region fixed effects:

$$Y_{it} = \beta D_{it} + \gamma X_{it} + \delta_t + \lambda_r + \epsilon_{it} \quad (3)$$

where D_{it} is defined as a time-invariant indicator for whether the respondent's first-born child is female, and λ_r denotes region fixed effects. These models are estimated on the full sample of parents with known first-child gender.

To test H_4, which posits heterogeneous effects by respondent gender, we extend the model with an interaction term:

$$Y_{it} = \beta_1 \mathrm{D}_i \times \mathrm{m}_i + \beta_2 \mathrm{D}_t + \beta_3 \mathrm{m}_i + \gamma X_{it} + \delta_t + \epsilon_{it} \quad (4)$$

where m_i is a binary indicator equal to one for male respondents, and the remaining variables are specified as stated above. The coefficient β_1 captures the differential effect of having daughters among men relative to women.

For household-level behavioral outcomes (H_3), we apply analogous specifications at the household level, restricting the sample to cohabiting biological parents observed in the same survey year. Outcome variables are framed as the father's share of total household labor or income. When both partners respond, household fixed effects are included where applicable.

All regressions are estimated using ordinary least squares (OLS). Identification relies on the assumption that, conditional on covariates, fixed effects, and survey year, the gender of the first-born child is as good as randomly assigned.

Required Sample Size. While the full dataset comprises approximately 90,000 observations, many of the outcome variables were administered only to subsamples, and fewer than 50% of the German population live in families with children or in households that include children.[2] As a result, the effective sample size for each specification is substantially smaller and varies by outcome availability, respondent type, and household composition.

[2]https://www.demografie-portal.de/DE/Fakten/lebensformen.html

To determine the minimum sample size required for adequately powered estimation, we conduct a multiple regression power calculation assuming an effect size of $f = 0.03$, five covariates, and a target power of 0.9. Under these assumptions, approximately 550 observations are needed to detect an effect at the 5% significance level ($\alpha = 0.05$). This threshold informs our empirical strategy: when the number of individuals meeting the criteria for difference-in-differences estimation—namely, being observed both before and after the birth of the first child—falls below 550, we switch to a pooled cross-sectional specification using all available observations with known first-child gender. Specifications with fewer than 550 complete observations are excluded from analysis.

This threshold is justified by the expected effect sizes in the literature. Empirical studies suggest that the effect of having daughters on preferences for gender equality is small—typically less than 3% of a standard deviation (Yang et al., 2025). If this range characterizes the true effect in our context, type II errors (false negatives) remain likely even near the 550-observation threshold. The switching rule therefore balances the need for internal validity in panel-based designs with the statistical power offered by larger pooled samples.

4.1 Statistical Judgment and Inference Criteria

All empirical analyses will be conducted conditional on the feasibility of estimation, as defined by sample size and outcome availability. Specifications that do not meet this threshold will be excluded from estimation and flagged as infeasible due to limited statistical power. This rule prevents the reporting of unstable or non-informative estimates and follows established guidance for credible pre-analysis planning.

All hypothesis tests will use two-sided p-values. We distinguish three levels of inferential strength:

- **Significant Evidence:** $p < 0.05$. Interpreted as statistically significant and consistent with the stated hypothesis, conditional on identification assumptions.
- **Suggestive Evidence:** $0.05 \leq p < 0.10$. Reported transparently as indicative of a possible effect, but not sufficient for conventional inference.
- **Non-significant:** $p \geq 0.10$. Reported with confidence intervals and interpreted in terms of magnitude and precision, without asserting evidentiary support.

Throughout, we prioritize the interpretation of effect sizes, uncertainty intervals, and robustness across specifications over dichotomous classifications of significance. Standardized coefficients will be reported where appropriate to enable comparisons across outcomes. All inferential conventions and decision thresholds are pre-specified to reduce the risk of selective reporting.

These practices are intended to maintain transparency and analytical discipline while allowing for reasonable interpretation under the constraints of observational data.

4.2 Robustness and Possible Limitations

Our identification strategy relies on the assumption that the biological sex of the first-born child is exogenous. This premise is well-supported by demographic and political behavior research, which finds that child sex at first birth is statistically random in contexts without prenatal sex selection or systematic gender preference (Washington, 2008; Dahl and Moretti, 2008). In the Germany during the observation window (2000–2020), there is no evidence of sex-selective abortion, and sex ratios at birth align with natural biological variation. Under these conditions, the gender of the first-born child provides a credible source of exogenous variation for estimating causal effects on parental attitudes and behavior.

To mitigate threats related to endogenous fertility timing, stopping behavior, or differential survey attrition around childbirth, we impose the following additional sampling restrictions: We limit the analysis to biological parents of first-born children with known and binary-coded gender and further restrict panel-based specifications to individuals observed both before and after the child's birth. For household-level analyses, we additionally condition on cohabitation and simultaneous survey participation of both biological parents.

While this approach improves internal validity, it limits external validity in several respects. First, the estimates should be interpreted as local average treatment effects for individuals in stable, cohabiting biological-parent dyads with children born during the panel window. These effects may not generalize to parents in second or subsequent marriages or to non-cohabiting parents. Relationship dynamics in higher-order partnerships may differ systematically due to accumulated life experiences or selective processes following relationship dissolution. For instance, parents in remarriages may adjust gender roles based on prior failures or negotiated expectations, leading to different baseline behaviors and treatment effects.

Second, the restriction to couples who remain cohabiting and survey-active may induce selection bias. Continued co-residence could reflect relationship quality, shared values, or codependence, all of which may be correlated with the outcomes of interest in either direction. This raises the possibility that observed associations reflect both the causal effect of child gender and unobserved characteristics associated with family stability. As a result, the sample is not representative of all parents, and caution is warranted in extrapolating the findings to broader populations.

Third, because the primary design leverages within-individual variation in a difference-in-differences framework, we assume that untreated potential outcomes would have evolved similarly across treatment groups. In the absence of formal pre-trend diagnostics, this assumption cannot be empirically verified within the current analysis.

References

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